INCH-POUND

MIL-DTL-13777/4C
13 June 2011
SUPERSEDING
MIL-C-13777/4B
15 September 1969

DETAIL SPECIFICATION SHEET

# CABLE, SPECIAL PURPOSE, ELECTRICAL 21, 22, 24, 26, 27, 28, AND 32 CONDUCTORS

Inactive for new design after 1 July 1998.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-13777.

REQUIREMENTS:

Dimensions and configuration: See applicable figure and design data for the following cable types specified in table I. Design data is specified in table II.

TABLE I. Cable types.

| | | | |
|---|---|---|---|
| 211055 | S221385 | 241215 | 280805 |
| 211115S | 240966 | 261065S | 321355 |
| 211345 | 241215S | 271100S | |

In any conflict between the design data and the applicable figure, the design data shall govern.



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:18Z
Check the source to verify that this is the current version before use.

TABLE II. Design data.

| | Design data | | | | |
|---|---|---|---|---|---|
| Type designation | | 211055 | | 211115S | |
| Figure number | | 1 | | 2 | |
| Total wires | | 21 | | 21 | |
| Number of conductors & AWG # | 15/#18 | 6/#12 | 5/#18S | 10/#18 | 6/#12 |
| Insulation | | | | | |
| Minimum average thickness | 0.015” | | 0.020” | 0.015” | 0.020” |
| Spark test voltage | 3000 | | 4000 | 3000 | 4000 |
| Inspection test voltage | 1500 | | 2000 | 1500 | 2000 |
| Cabling | | | | | |
| Layer 1 | | Filler | | Filler | |
| Layer 2 | | | | | |
| (a) Number of wires | | 6 | | 6 | |
| (b) AWG # | | #12 | | #12 | |
| (c) Maximum lay | | 4.00” | | 4.00” | |
| Layer 3 | | | | | |
| (a) Number of wires | | 15 | | 5 | 10 |
| (b) AWG # | | #18 | | #18S | #18 |
| (c) Maximum lay | | 7.00” | | 7.00” | |
| Sheath | | | | | |
| Number of layers | | 2 | | 2 | |
| Total thickness minimum | | 0.120” | | 0.120” | |
| Minimum OD cable | | 1.040” | | 1.100” | |
| Maximum OD cable | | 1.070” | | 1.130” | |
| Maximum weight (lbs./1000ft.) | | 708 | | 791 | |

See notes at end of table.



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:18Z
Check the source to verify that this is the current version before use.

TABLE II. Design data – Continued.

| | Design data | | | | | |
|---|---|---|---|---|---|---|
| Type designation | | 211345 | | | S221385 | |
| Figure number | | 1 | | | 3 | |
| Total wires | | 21 | | | 22 | |
| Number of conductors & AWG # | 15/#18 | | 6/#9 | 3/#8 | 4/#12 | 14/#16 |
| Insulation | | | | | | |
| Minimum average thickness | 0.015” | | 0.030” | 0.060” | 0.020” | 0.015” |
| Spark test voltage | 3000 | | 4000 | 7500 | 4000 | 3000 |
| Inspection test voltage | 1500 | | 2000 | 5000 | 2000 | 1500 |
| Cabling | | | | | | |
| Layer 1 | | Filler | | | Filler | |
| Layer 2 | | | | | | |
| (a) Number of wires | | 6 | | | 6 | |
| (b) AWG # | | #9 | | 3/#8 | | 3/#16 |
| (c) Maximum lay | | 6.00” | | | | |
| Layer 3 | | | | | | |
| (a) Number of wires | | 15 | | | 16 | |
| (b) AWG # | | #18 | | 4/#12 | | 12/#16 |
| (c) Maximum lay | | 9.00” | | | 6.00” | |
| Sheath | | | | | | |
| Number of layers | | 2 | | | 2 | |
| Total thickness minimum | | 0.156” | | | 0.156” | |
| Minimum OD cable | | 1.330” | | | 1.360” | |
| Maximum OD cable | | 1.360” | | | | |
| Maximum weight (lbs./1000ft.) | | 980 | | | | |

See notes at end of table.



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:18Z
Check the source to verify that this is the current version before use.

TABLE II. Design data – Continued.

| | Design data | | | |
|---|---|---|---|---|
| Type designation | 240966 | | 241215S | |
| Figure number | 4 | | 5 | |
| Total wires | 24 | | 24 | |
| Number of conductors & AWG # | 24/#18 | 12/#18S[1/] | | 12/#18S[2/] |
| Insulation | | | | |
| Minimum average thickness | 0.015” | | 0.015” | |
| Spark test voltage | 3000 | | 3000 | |
| Inspection test voltage | 1500 | | 1500 | |
| Cabling | | | | |
| Layer 1 | | | | |
| (a) Number of wires | 2 | | 4 | |
| (b) AWG # | #18 | | #18S[3/] | |
| (c) Maximum lay | | | 3.50” | |
| Layer 2 | | | | |
| (a) Number of wires | 8 | 8 | | 12 |
| (b) AWG # | #18 | #18S[4/] | | #18S[5/] |
| (c) Maximum lay | | | 7.00” | |
| Layer 3 | | | | |
| (a) Number of wires | 14 | | | |
| (b) AWG # | #18 | | | |
| (c) Maximum lay | 4.5” | | | |
| Sheath | | | | |
| Number of layers | 2 | | 2 | |
| Total thickness minimum | 0.125” | | 0.100” | |
| Minimum OD cable | 0.951” | | 1.200” | |
| Maximum OD cable | 0.981” | | 1.230” | |

See notes at end of table.



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:18Z
Check the source to verify that this is the current version before use.

TABLE II. Design data – Continued.

| | Design data | | | | | |
|---|---|---|---|---|---|---|
| Type designation | | 241215 | | | 261065S | |
| Figure number | | 6 | | | 7 | |
| Total wires | | 24 | | | 26 | |
| Number of conductors & AWG # | 5/#18 | 19/#12 | 5/#18S | 6/#18 | 12/#13 | 3/#12 |
| Insulation | | | | | | |
| Minimum average thickness | 0.015” | | 0.020” | 0.015” | | 0.018” |
| Spark test voltage | 3000 | | 4000 | 3000 | | 4000 |
| Inspection test voltage | 1500 | | 2000 | 1500 | | 1800 |
| Cabling | | | | | | |
| Layer 1 | | Filler | | | | |
| (a) Number of wires | | | | | 2 | |
| (b) AWG # | | | | | #18S[6/] | |
| (c) Maximum lay | | | | | | |
| Layer 2 | | | | | | |
| (a) Number of wires | | 5 | 3 | | | 6 |
| (b) AWG # | | #18 | #18S | | | #13 |
| (c) Maximum lay | | | | | 5.00” | |
| Layer 3 | | | | | | |
| (a) Number of wires | | 7 | 6 | | 6 | 3 |
| (b) AWG # | | #12 | #18 | | #13 | #12 |
| (c) Maximum lay | | | | | 6.00” | |
| Layer 4 | | | | | | |
| (a) Number of wires | | 12 | | | | |
| (b) AWG # | | #12 | | | | |
| (c) Maximum lay | | | | | | |
| Sheath | | | | | | |
| Number of layers | | 2 | | | 1 | |
| Total thickness minimum | | 0.098” | | | 0.080” | |
| Minimum OD cable | | 1.190” | | | 1.050” | |
| Maximum OD cable | | 1.240” | | | 1.080” | |
| Maximum weight (lbs./1000ft.) | | | | | 750 | |

See notes at end of table.



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:18Z
Check the source to verify that this is the current version before use.

TABLE II. Design data – Continued.

| | Design data | |
|---|---|---|
| Type designation | 271100S | 280805 |
| Figure number | 8 | 9 |
| Total wires | 27 | 28 |
| Number of conductors & AWG # | 27/#18 | #20 |
| Insulation | | |
| Minimum average thickness | 0.012” | 0.015” |
| Spark test voltage | 3000 | 3000 |
| Inspection test voltage | 1500 | 1500 |
| Cabling | | |
| Layer 1 | | |
| (a) Number of wires | 3 | 4 |
| (b) AWG # | #18 | #20 |
| (c) Maximum lay | | |
| Layer 2 | | |
| (a) Number of wires | 9 | 9 |
| (b) AWG # | #18 | #20 |
| (c) Maximum lay | 5.00” | |
| Layer 3 | | |
| (a) Number of wires | 15 | 15 |
| (b) AWG # | #18S | #20 |
| (c) Maximum lay | | |
| Sheath | | |
| Number of layers | 2 | 1 |
| Total thickness minimum | 0.109” | 0.080” |
| Minimum OD cable | 1.085” | 0.780” |
| Maximum OD cable | 1.115” | 0.830” |
| Maximum weight (lbs./1000ft.) | 780 | |

See notes at end of table.



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:18Z
Check the source to verify that this is the current version before use.

TABLE II. Design data – Continued.

| | Design data | |
|---|---|---|
| Type designation | 321355 | |
| Figure number | 10 | |
| Total wires | 48 | |
| Number of conductors & AWG7/ # | | |
| Insulation | | |
| Minimum average thickness | 0.015” | |
| Spark test voltage | 3000 | |
| Inspection test voltage | 1500 | |
| Cabling | | |
| Layer 1 | Filler | |
| Layer 2 | | |
| (a) Number of wires | 5 | |
| (b) AWG # | #16 | #20 |
| (c) Maximum lay | 2.0” | |
| Layer 3 | | |
| (a) Number of wires | 11 | |
| (b) AWG # | #16 | #20 |
| (c) Maximum lay | 6.5” | |
| Sheath | | |
| Number of layers | 2 | |
| Total thickness minimum | 0.100” | |
| Minimum OD cable | 1.325” | |
| Maximum OD cable | 1.385” | |
| Maximum weight (lbs./1000ft.) | 945 | |

1/ Shielded pairs
2/ Shielded triplets
3/ 2 Shielded pairs
4/ 4 Shielded pairs
5/ 4 Shielded triplets
6/ Shielded pairs AWG#18 with 3” Max lay twist
7/ 32 #16 shielded pairs w/16 #20 bare



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:18Z
Check the source to verify that this is the current version before use.

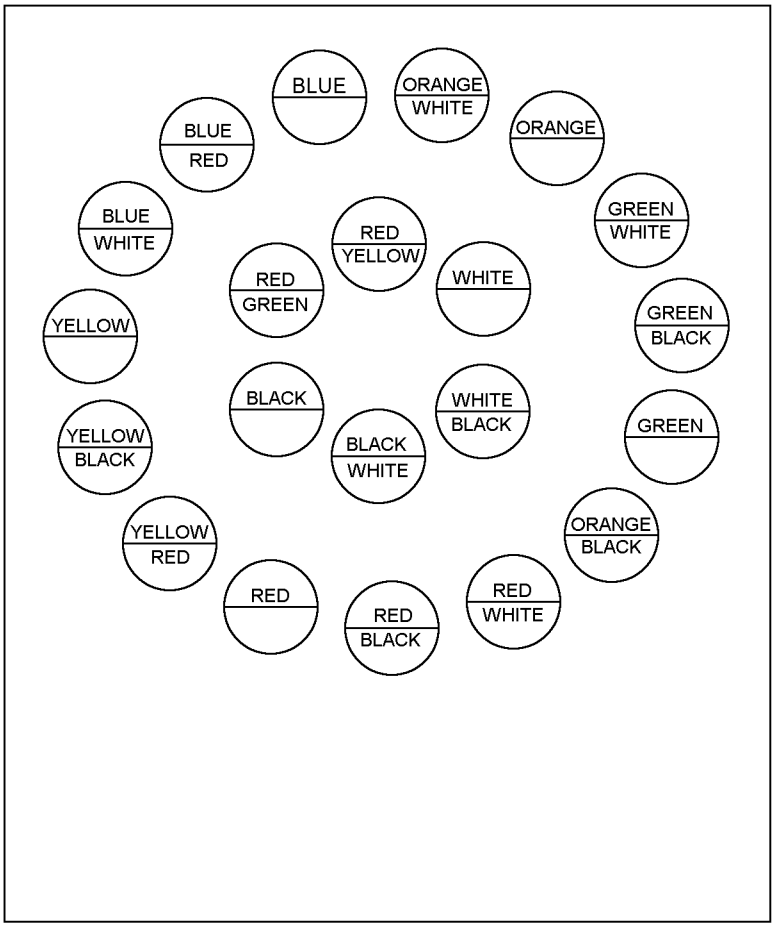
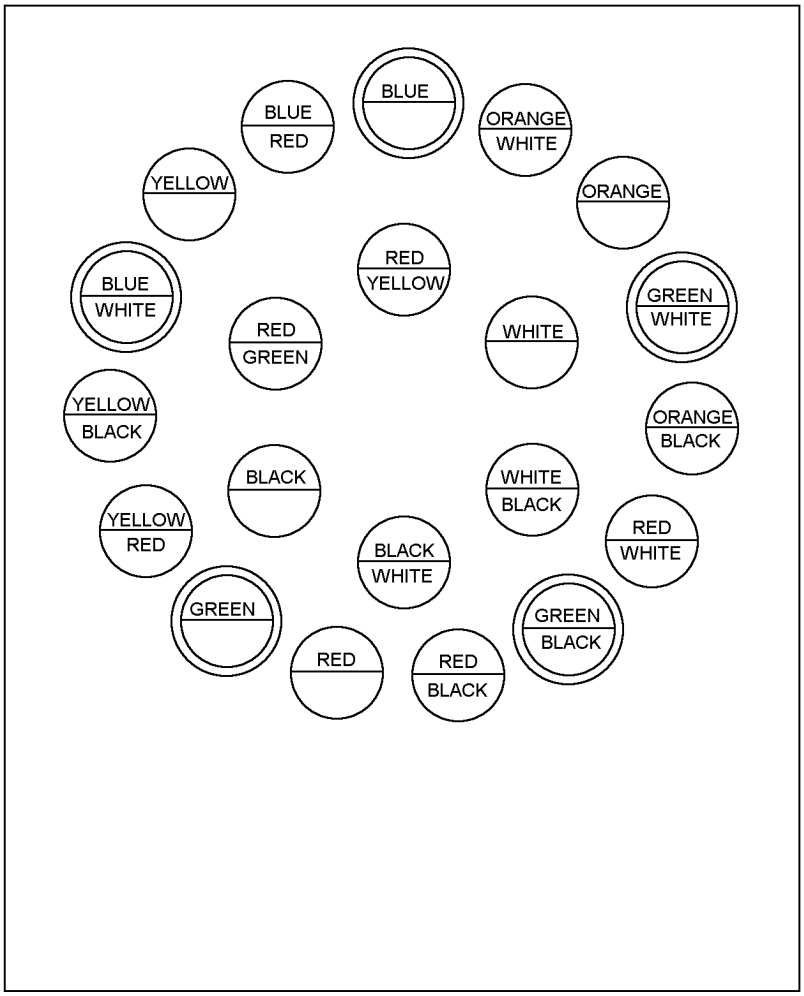



FIGURE 1. Cable types (211055, 211345).



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:18Z
Check the source to verify that this is the current version before use.




NOTE: Double circle indicates shielded conductors.

FIGURE 2. Cable type (211115S).



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:18Z
Check the source to verify that this is the current version before use.

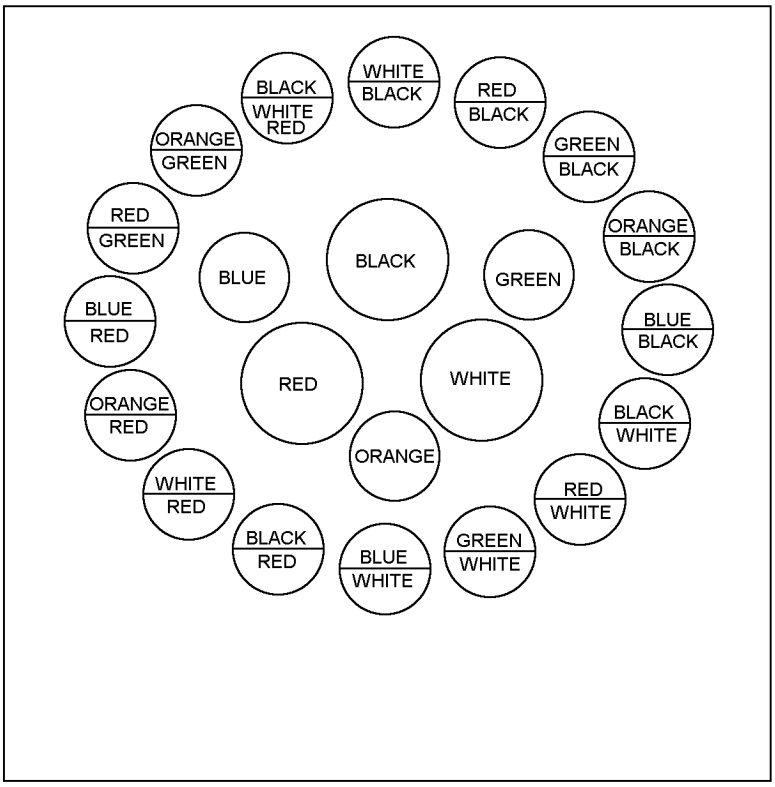



FIGURE 3. Cable type (S221385).



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:18Z
Check the source to verify that this is the current version before use.

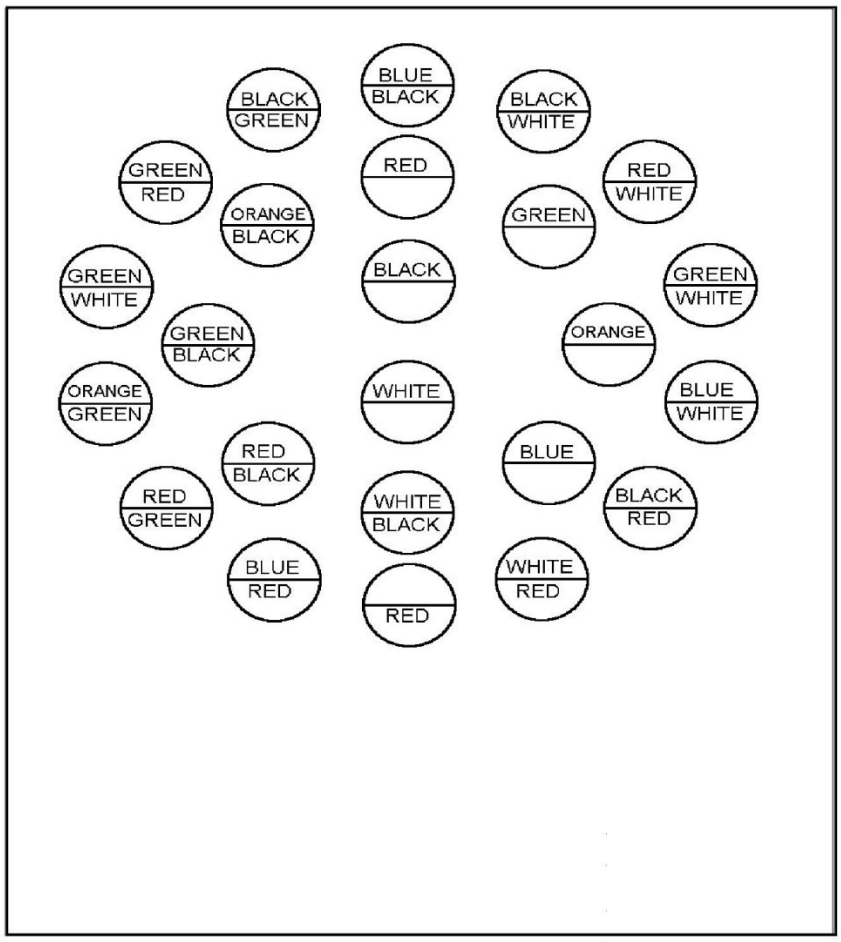
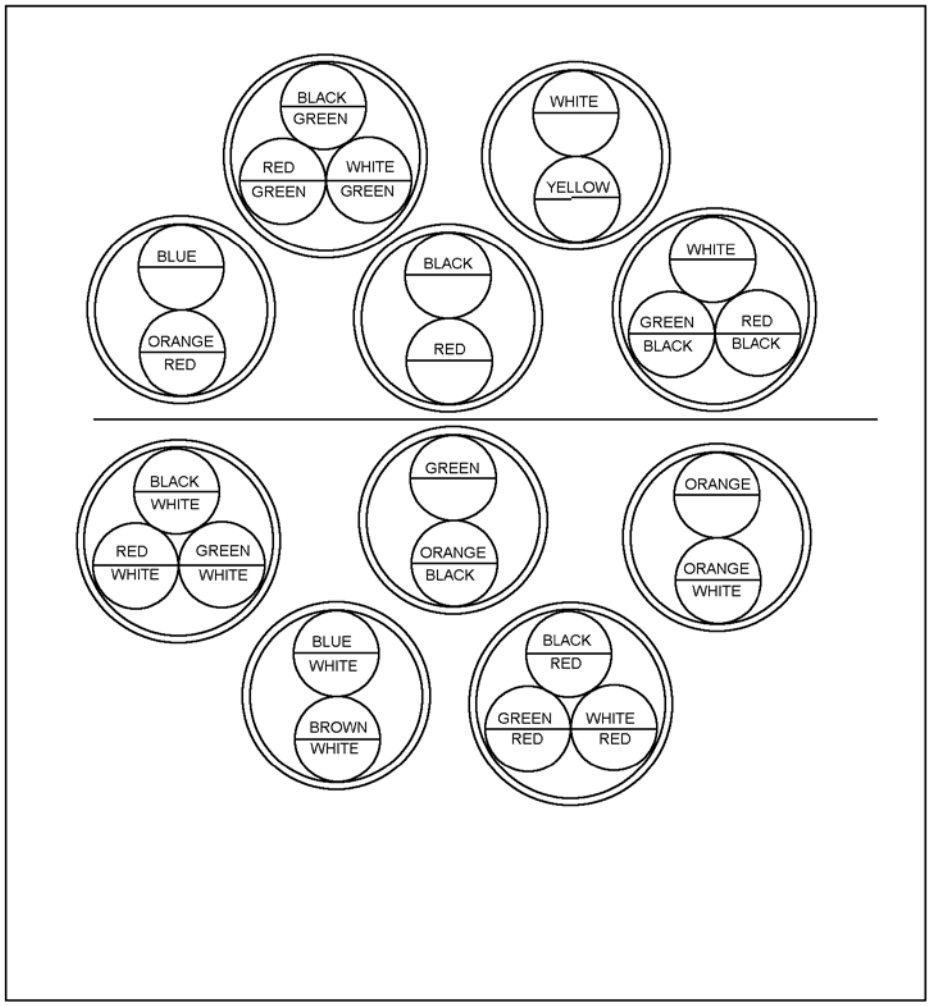



FIGURE 4. Cable type(240966).



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:18Z
Check the source to verify that this is the current version before use.




NOTE: Double shield indicates shielded conductors.

FIGURE 5. Cable type (241215S).



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:18Z
Check the source to verify that this is the current version before use.

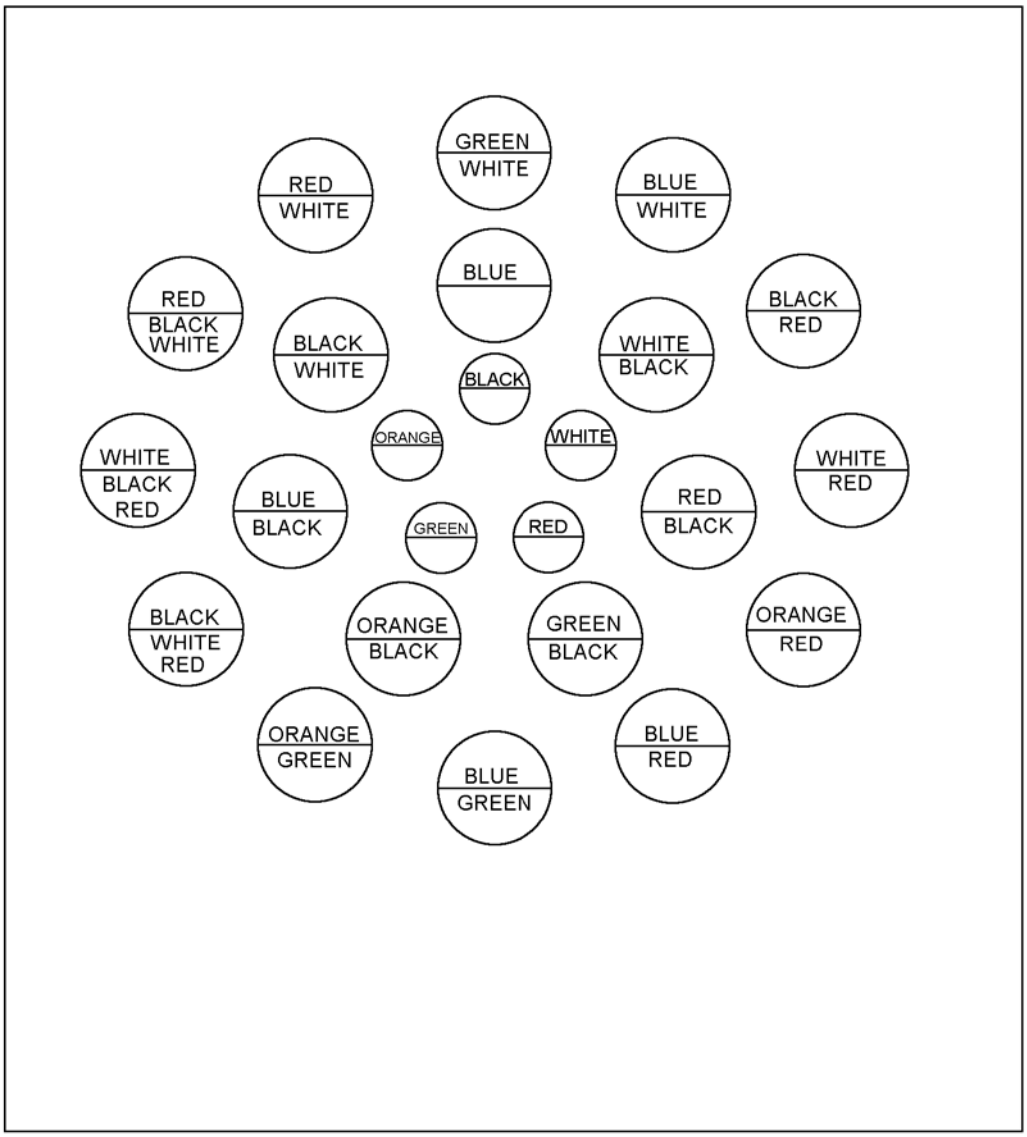
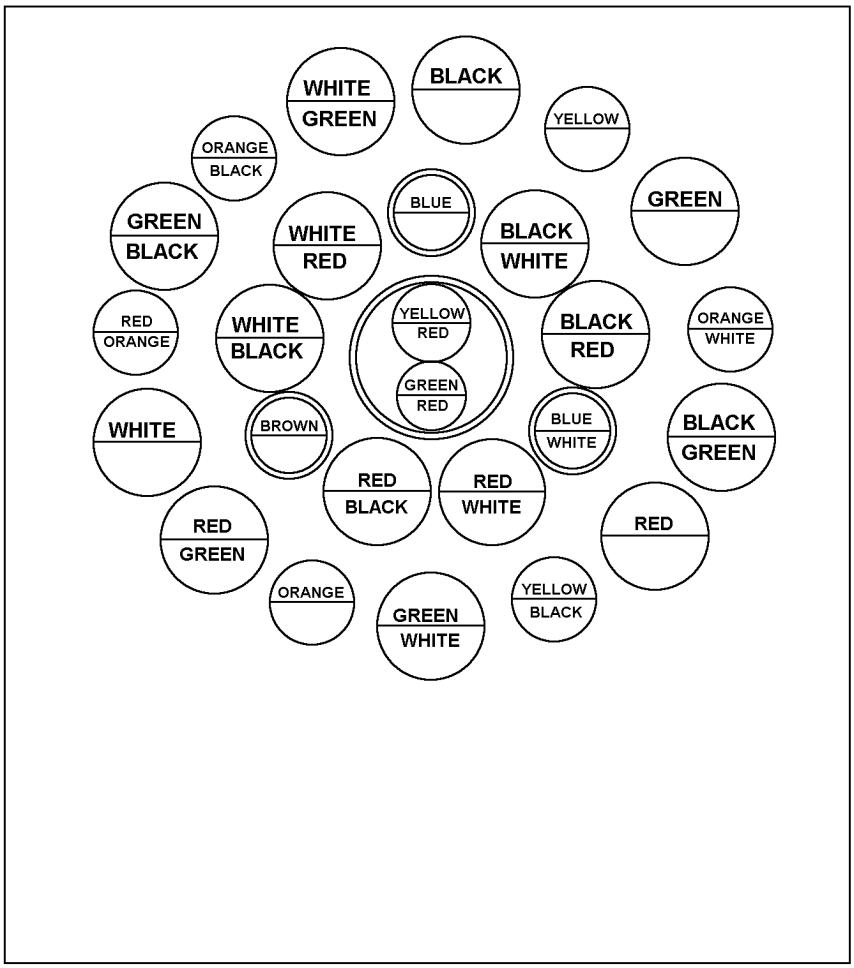



FIGURE 6. Cable type (241215).



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:18Z
Check the source to verify that this is the current version before use.

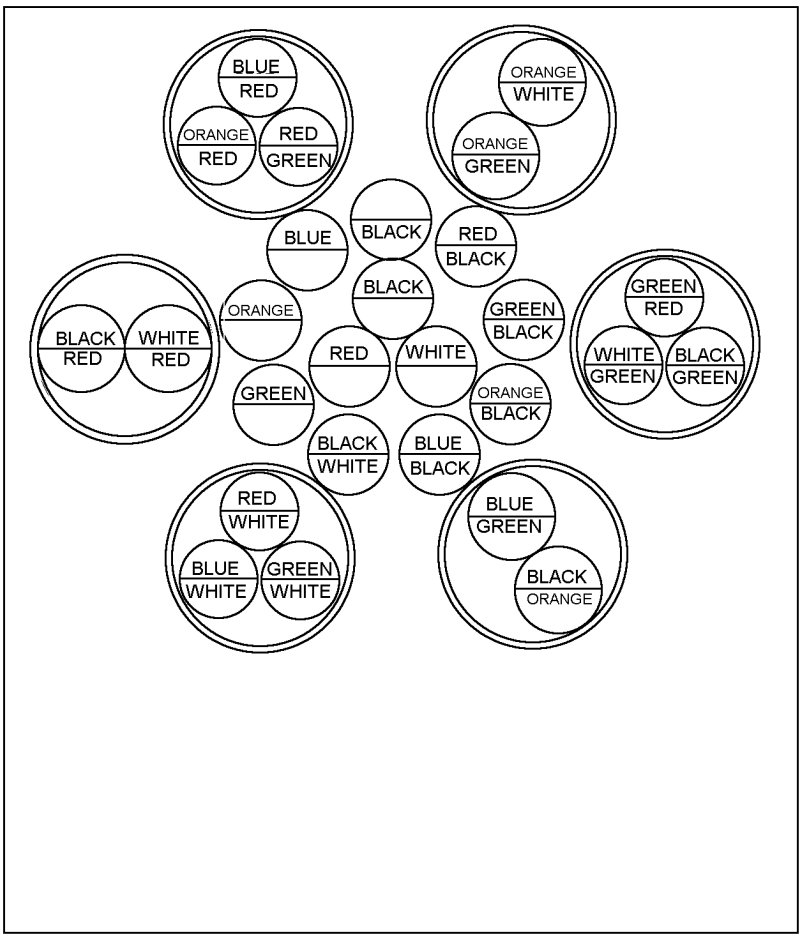



NOTE: Double circle indicates shielded conductors.

FIGURE 7. Cable type (261065S).



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:18Z
Check the source to verify that this is the current version before use.




NOTE: Double circle indicates shielded conductors.

FIGURE 8. Cable type (271100S).



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:18Z
Check the source to verify that this is the current version before use.

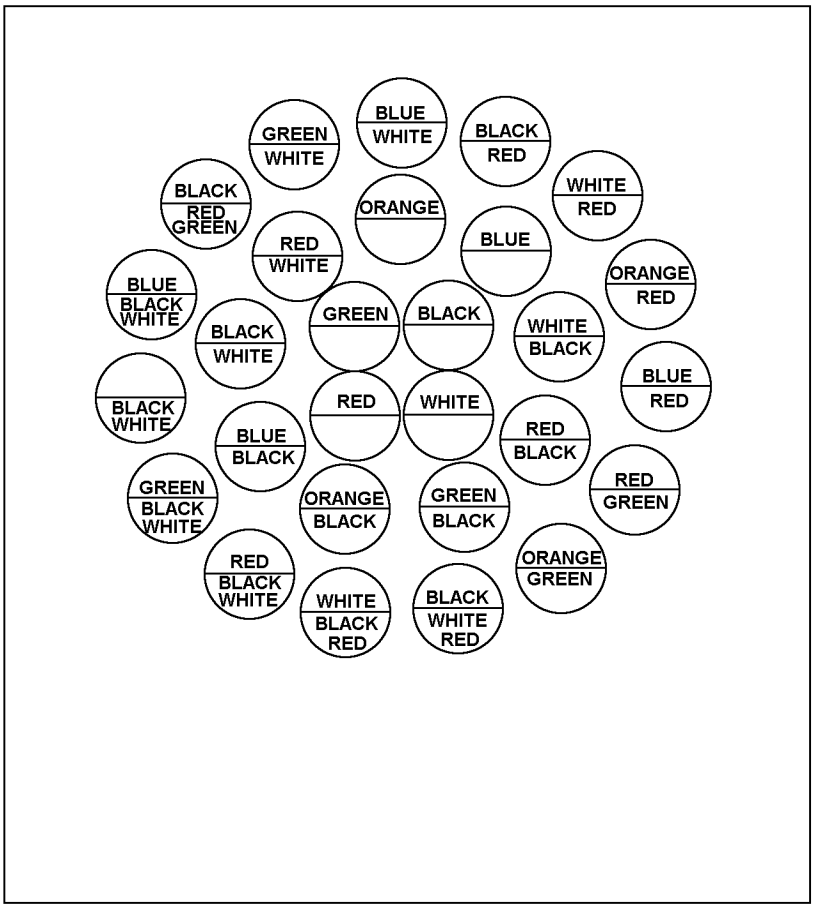



FIGURE 9. Cable type (280805).



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:18Z
Check the source to verify that this is the current version before use.

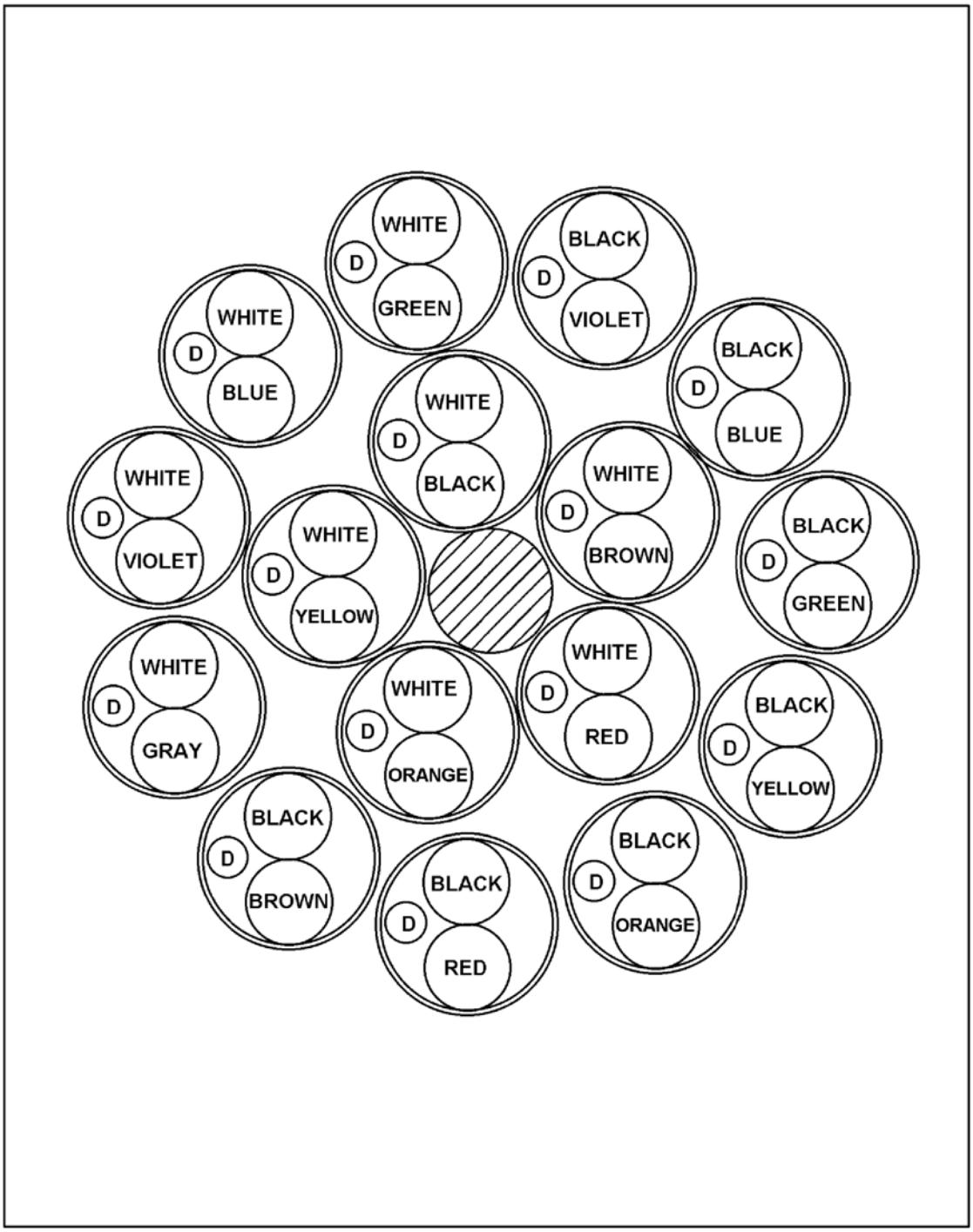



FIGURE 10. Cable type (431355).



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:18Z
Check the source to verify that this is the current version before use.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issues due to the extensiveness of the changes.

Referenced documents. This document references MIL-DTL-13777.

CONCLUDING MATERIAL

Custodians:
Army – AV
Navy – SH
DLA - CC

Preparing activity:
DLA - CC

(Project 6145-2011-010)

Review activities:
Army – AR, CR, MI
Air Force – 85

Note: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information using the Assist Online database at https://assist.daps.dla.mil.



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T02:18Z
Check the source to verify that this is the current version before use.